Sample data for selling long term bank loans

To help manage the dataset and better understand it I have cleaned the data and performed regression to help recognise trends and see which variables will give the higher probability of a successful call to sell long term bank loans.

There are 21 variables in the dataset and 4100 observations. As shown in fig.1.

```
> dim(sampledata)
[1] 4100   21
```

Fig.1. dim function.

As shown below in Fig.2. I have also viewed the types of variables and this allows me to see I have three types of variables, integer, numeric and character variables. it also shows me the variables names and what observations are in the variables.

```
> str(sampledata)
'data.frame':   4100 obs. of  21 variables:
 $ age           : int  30 39 25 38 47 32 32 41 31 35 ...
 $ job           : chr  "blue-collar" "services" "services" "services" ...
 $ marital       : chr  "married" "single" "married" "married" ...
 $ k             : chr  "basic.9y" "high.school" "high.school" "basic.9y" ...
 $ default       : chr  "no" "no" "no" "no" ...
 $ housing       : chr  "yes" "no" "yes" "unknown" ...
 $ loan          : chr  "no" "no" "no" "unknown" ...
 $ contact       : chr  "cellular" "telephone" "telephone" "telephone" ...
 $ month         : chr  "may" "may" "jun" "jun" ...
 $ day_of_week   : chr  "fri" "fri" "wed" "fri" ...
 $ duration      : int  487 346 227 17 58 128 290 44 68 170 ...
 $ campaign      : int  2 4 1 3 1 3 4 2 1 1 ...
 $ pdays         : int  999 999 999 999 999 999 999 999 999 999 ...
 $ previous      : int  0 0 0 0 0 2 0 0 1 0 ...
 $ poutcome      : chr  "nonexistent" "nonexistent" "nonexistent" "nonexistent" ...
 $ emp.var.rate  : num  -1.8 1.1 1.4 1.4 -0.1 -1.1 -1.1 -0.1 -0.1 1.1 ...
 $ cons.price.idx: num  92.9 94 94.5 94.5 93.2 ...
 $ cons.conf.idx : num  -46.2 -36.4 -41.8 -41.8 -42 -37.5 -37.5 -42 -42 -36.4 ...
 $ euribor3m     : num  1.31 4.86 4.96 4.96 4.19 ...
 $ nr.employed   : num  5099 5191 5228 5228 5196 ...
 $ y             : chr  "no" "no" "no" "no" ...
```

Fig.2. variable types

The dependent variables are variables that rely on the independent variables to change or occur before they can have a result.

The independent variables are variables that can be manipulated and changed to be able to change the outcome so for example as we are trying to sell long term bank loans, if we changed the month we contact a person in for example we could change the outcome of the success of that call.

Below I have organised the variables into the dependent and independent variables in table.1.

Table.1. dependent and independent variables.

Dependent Variables	**Independent Variables**
age	Contact type
job	Last month they were contacted in
Education	Last day they were contacted
Marital status	Number of contacts during the campaign
Has credit in default	number of days since last contact from a previous campaign
Has housing loan	Number of contacts performed before this campaign
Has personal loan	Outcome of previous marketing campaign
	Employment variation rate
	Consumer price index
	consumer confidence index
	Euribor 3 month rate
	Number of employees
	Duration of the call

The variables are related to each other in a sense that one variable has an affect on the other, For example the last day a person was contacted relates to the last month variable so this tells us the day in the month that person was last contacted. The number of contacts during the campaign also relates to the number of contacts performed during this campaign so this will tell us if this campaign was more successful for that person than the last campaign. The duration of a call can also relate to the contact type so we can see if the duration of a call is longer with a particular contact type and the longer the call might indicate more success from that call. Housing and personal loan relate to each other as they are both loan types.

EDA is the process of investigating data sets to help discover patterns, anomalies and helps to form a hypothesis based on our understanding of the dataset. We use EDA to generate summary statistics for the data and create various graphs to help us to understand the data better.

Below I have cleaned the data by checking for missing data, removing missing data, and changing some data to make it easier to read.

In Fig.3. below I have used the code na.omit to remove any missing data from the dataset and I also used unique to remove any duplicates in the dataset.

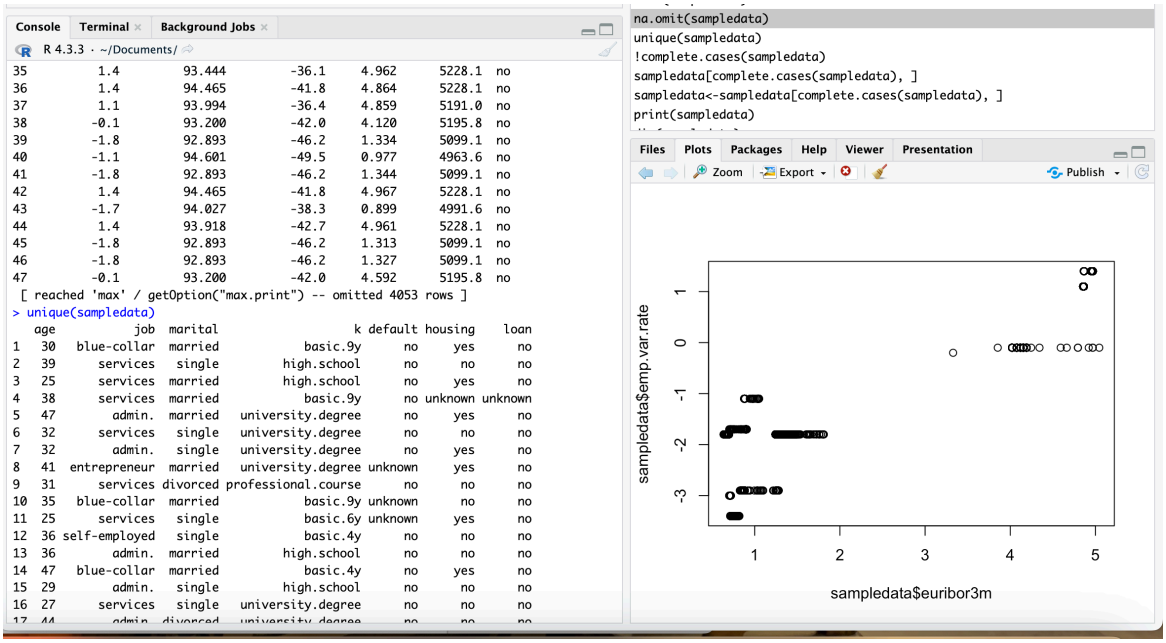


Fig.3. cleaning the data

In Fig.4. I used the code complete.cases to check which data had any missing data within the dataset. Having clean data makes our finds within the data more reliable and easier to read without creating charts with inapplicable data.

```
 [ reached 'max' / getOption("max.print") -- omitted 4053 rows ]
> !complete.cases(sampledata)
   [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [37] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [49] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [73] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
  [97] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [109] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [121] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [133] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [145] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [157] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [169] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [181] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [193] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [205] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [217] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [229] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [241] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [253] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
 [265] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

Fig.4. cleaning missing data

In Fig.5. I used the code below to look at all the complete data. This was so I could see all the data I was working with and to ensure it was all complete data.

```
> sampledata[complete.cases(sampledata), ]
   age           job  marital                   k default housing    loan
1   30   blue-collar  married           basic.9y      no     yes      no
2   39      services   single        high.school      no      no      no
3   25      services  married        high.school      no     yes      no
4   38      services  married           basic.9y      no unknown unknown
5   47        admin.  married   university.degree      no     yes      no
6   32      services   single   university.degree      no      no      no
7   32        admin.   single   university.degree      no     yes      no
8   41  entrepreneur  married   university.degree unknown     yes      no
9   31      services divorced professional.course      no      no      no
10  35   blue-collar  married           basic.9y unknown      no      no
11  25      services   single           basic.6y unknown     yes      no
12  36 self-employed   single           basic.4y      no      no      no
13  36        admin.  married        high.school      no      no      no
14  47   blue-collar  married           basic.4y      no     yes      no
15  29        admin.   single        high.school      no      no      no
16  27      services   single   university.degree      no      no      no
17  44        admin. divorced   university.degree      no      no      no
18  46        admin. divorced   university.degree      no     yes      no
19  45  entrepreneur  married   university.degree unknown     yes     yes
20  50   blue-collar  married           basic.4y      no      no     yes
21  55      services  married           basic.6y unknown     yes      no
22  39    technician divorced        high.school      no      no      no
23  29    technician   single   university.degree      no     yes     yes
24  40    management  married        high.school      no      no     yes
25  44    technician  married professional.course unknown     yes      no
26  38    technician  married professional.course      no     yes      no
27  36    technician divorced professional.course      no      no      no
28  28   blue-collar  married           basic.6y unknown      no      no
```

Fig.5. checking all the complete data.

In Fig.6. I completed the action of cleaning the data to work with.

```
> sampledata<-sampledata[complete.cases(sampledata), ]
> print(sampledata)
   age           job  marital                  k default housing    loan
1   30   blue-collar  married           basic.9y      no     yes      no
2   39      services   single        high.school      no      no      no
3   25      services  married        high.school      no     yes      no
4   38      services  married           basic.9y      no unknown unknown
5   47        admin.  married   university.degree      no     yes      no
6   32      services   single   university.degree      no      no      no
7   32        admin.   single   university.degree      no     yes      no
8   41  entrepreneur  married   university.degree unknown     yes      no
9   31      services divorced professional.course      no      no      no
10  35   blue-collar  married           basic.9y unknown      no      no
11  25      services   single           basic.6y unknown     yes      no
12  36 self-employed   single           basic.4y      no      no      no
13  36        admin.  married        high.school      no      no      no
14  47   blue-collar  married           basic.4y      no     yes      no
15  29        admin.   single        high.school      no      no      no
16  27      services   single   university.degree      no      no      no
17  44        admin. divorced   university.degree      no      no      no
18  46        admin. divorced   university.degree      no     yes      no
19  45  entrepreneur  married   university.degree unknown     yes     yes
20  50   blue-collar  married           basic.4y      no      no     yes
21  55      services  married           basic.6y unknown     yes      no
22  39    technician divorced        high.school      no      no      no
23  29    technician   single   university.degree      no     yes     yes
24  40    management  married        high.school      no      no     yes
25  44    technician  married professional.course unknown     yes      no
26  38    technician  married professional.course      no     yes      no
27  36    technician divorced professional.course      no      no      no
28  28   blue-collar  married           basic.6y unknown      no      no
```

Fig.6. clean data.

Once the data is cleaned I can perform the function to create charts to be able to identify trends or patterns within the data and I can also see which variables have the most positive effect on having a successful call for selling long term bank loans.

First I perform the functions to create charts for the variables I think have the most effect on having successful calls as shown in Fig.7.

```
> ggplot(sampledata)+
+   geom_point(mapping=aes(emp.var.rate, y=k))
> ggplot(sampledata)+
+   geom_point(mapping=aes(x=emp.var.rate, y=k))
> ggplot(sampledata)+
+   geom_point(mapping=aes(x=age, y=emp.var.rate))
> ggplot(sampledata)+
+   geom_smooth(mapping=aes(x=age, y=emp.var.rate))
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
> ggplot(sampledata)+
+   geom_smooth(mapping=aes(x=cons.price.idx, y=emp.var.rate))
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
> ggplot(sampledata)+
+   geom_smooth(mapping=aes(x=cons.price.idx, y=euribor3m))
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

Fig.7. creating the charts

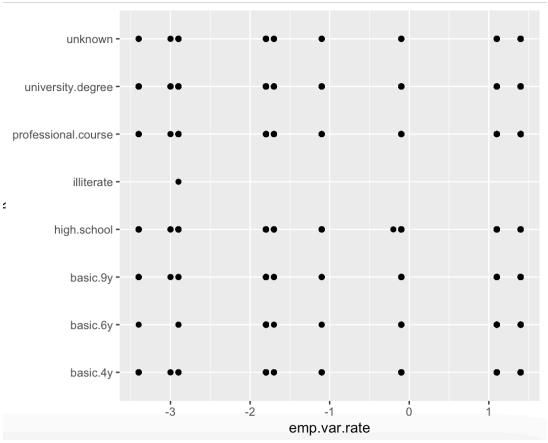


Fig.8. correlation between education and emp.var.rate.

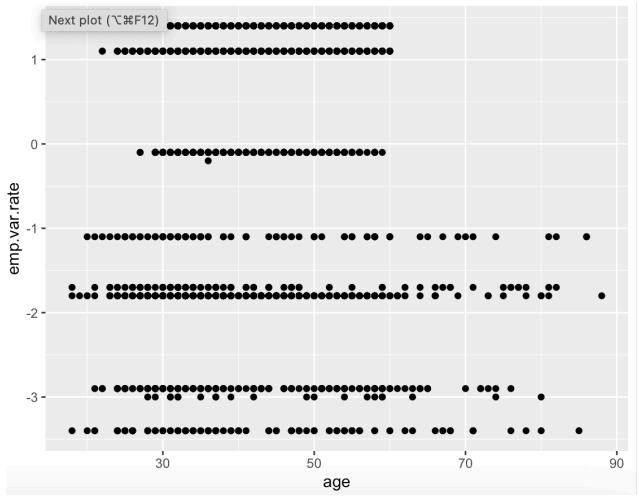


Fig.9. correlation between emp.var.rate and age.

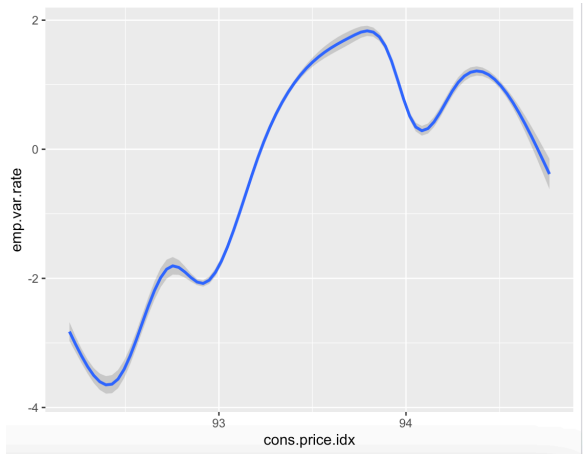


Fig.10. correlation between emp.var.rate and cons.price.idx

From this chart we can see as the emp.var.rate increases so does the cons.price.idx until it reaches about 93.5 and then it drops and then goes back up again.

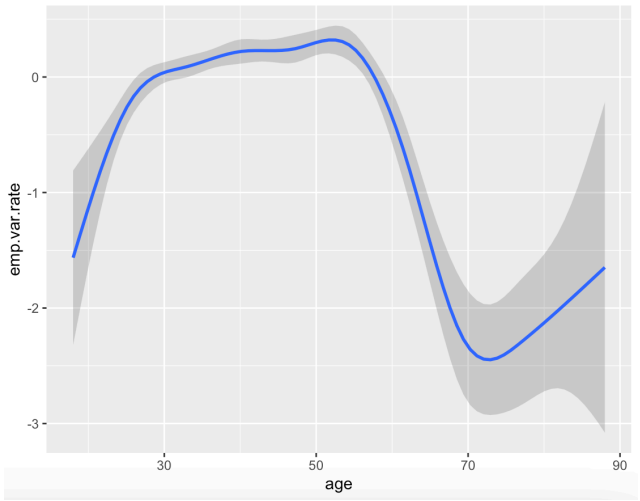


Fig.11. correlation between age and emp.var.rate.

This chart shows us as the age goes up so does the emp.var.rate until about 55 years old and then it has a big drop from about 60 years old to 70 years old and at 70 starts to rise slightly. This can help to determine which age range has the most success for buying long term bank loans.

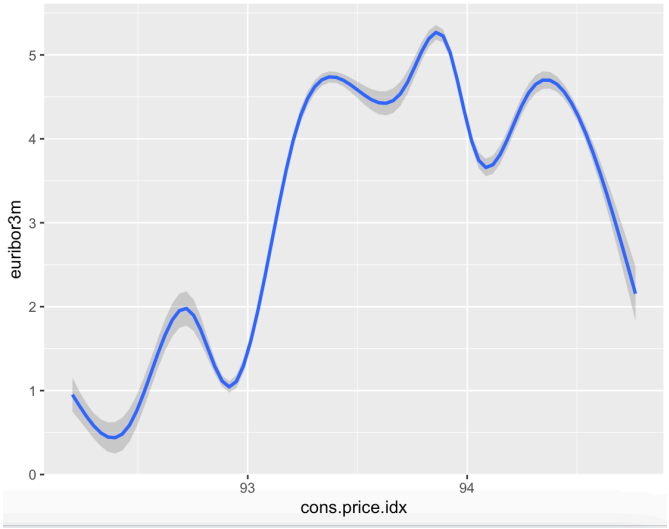


Fig.12. correlation between euribor3m and cons.price.idx.

This chart shows a fluctuation throughout the chart so as the cons.price.idx grows the euribor3m goes up and down with its peak just before cons.price.idx is 94.

This data has been gathered to help determine what can help in making successful calls to sell long term bank loans.

I chose these variables to look at because I think they are the most appropriate figures to see.
Euribor3m is the Euribor 3 month rate, this is an interest rate which a selection of banks lend each other and it has a maturity of 3 months. This has an effect on a person buying long term bank loans because the higher the rate the less likely a person is to accept it.
Cons.price.idx is weather or not a person has credit in default, this has an effect on a person buying long term bank loans because if they have credit in default it could mean they won't get accepted to buy a long term bank loan.
emp.var.rate is the employment variation rate and this means it can have a negative impact on clients purchasing. This has an effect on a person buying long term bank loans because large employment rate changes make it less likely for a person to buy a long term bank loan.
Age is how old a person is and can have an effect on a person buying a long term bank loan because circumstances at certain ages might mean they can't buy one or it means they are at an age of buying a nice car etc and will buy one.
K is education and this is the highest level of education that person has obtained. This has an effect because they have more awareness of the world and how it works and are educated more than others about it.
So as we can see from the charts mid age groups are more likely to buy long term bank loans as the employment variation rate reaches 1. As the employment variate rate also reaches 1 30-55 year olds have a high chance of buying a long term bank loan.

To create a regression model I first worked out the correlation between euribor3m and emp.var.rate. using the code shown in Fig.13. I used the cor function to work out the correlation and then plotted the data as shown in Fig.14.

```
> cor(sampledata$euribor3m, sampledata$emp.var.rate)
[1] 0.9703124
> plot(sampledata$euribor3m, sampledata$emp.var.rate)
> model<-lm(euribor3m~emp.var.rate, data=sampledata)
> model

Call:
lm(formula = euribor3m ~ emp.var.rate, data = sampledata)

Coefficients:
 (Intercept)  emp.var.rate
       3.530         1.076

> summary(model)

Call:
lm(formula = euribor3m ~ emp.var.rate, data = sampledata)

Residuals:
     Min       1Q   Median       3Q      Max
-1.46999 -0.17239 -0.07339  0.14346  1.62285

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.529763   0.006559   538.2   <2e-16 ***
emp.var.rate 1.076162   0.004190   256.8   <2e-16 ***
---
```

Fig.13. correlation between euribor3m and emp.var.rate.

I then plotted the correlation chart.

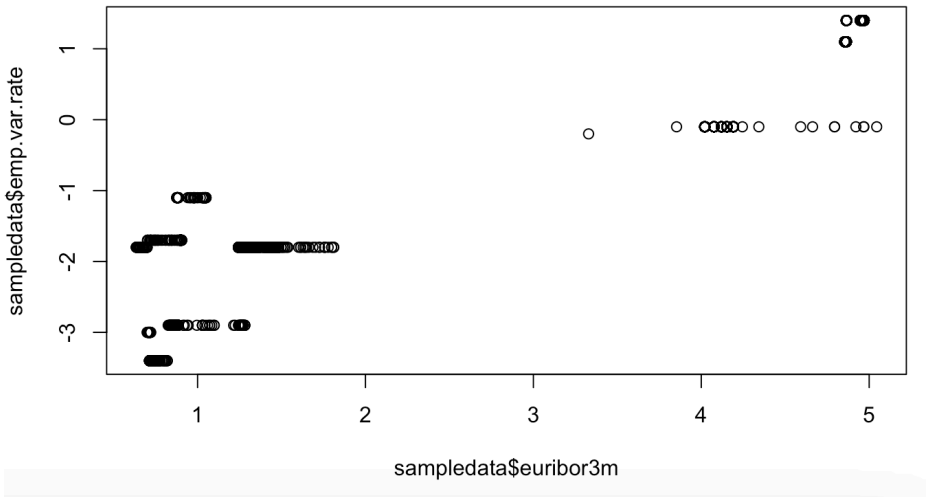


Fig.14. chart of correlation between emp.var.rate and euribor3m.

In Fig.15. I then worked out the standard deviation and R squared of the data.

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.529763   0.006559   538.2   <2e-16 ***
emp.var.rate 1.076162   0.004190   256.8   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4194 on 4098 degrees of freedom
Multiple R-squared:  0.9415,    Adjusted R-squared:  0.9415
F-statistic: 6.596e+04 on 1 and 4098 DF,  p-value: < 2.2e-16

> sigma(model)
[1] 0.4193571
> summary(model)$r.squared
[1] 0.9415062
> predict(model)
        1         2         3         4         5         6         7
1.5926716 4.7135418 5.0363905 5.0363905 3.4221472 2.3459851 2.3459851
        8         9        10        11        12        13        14
3.4221472 3.4221472 4.7135418 5.0363905 5.0363905 4.7135418 5.0363905
       15        16        17        18        19        20        21
1.5926716 5.0363905 5.0363905 5.0363905 5.0363905 5.0363905 5.0363905
       22        23        24        25        26        27        28
```

Fig.15. working out the SD and R squared.

I then used the predict function to predict some of the outcomes at certain values of the emp.var.rate as shown below in Fig.16.

```
> predict(model,data.frame(emp.var.rate=1))
       1
4.605926
> predict(model,data.frame(emp.var.rate=-2.5))
        1
0.8393581
>
> rangeRev<-data.frame(emp.var.rate<-c(1,2,3,4,5))
> predict(model,rangeRev)
       1        2        3        4        5
4.605926 5.682088 6.758250 7.834412 8.910574
```

Fig.16. predictions.

Conclusion:

From the data I gathered and looking at the charts I believe contacting customers who are age range 35-50 years old, are educated to university degree, the employment variation rate is at work as worked out in the linear regression calculation stating the linear regression calculation was 1.076, a consumer price index of 94 and the euribor3m is between 3-4 as worked out in the linear regression calculations showing 3.530. These ranges seem to have the most success for buying long term bank loans.

Looking at the data to find the most suitable customers with these figures would give the most success of selling long term bank loans.

References:

Datacamp. (2023). Random forest classification with scikit-learn. The dataset. Available at: https://www.datacamp.com/tutorial/random-forests-classifier-python# [date accessed 02nd May 2024].

Global rates. Euroibor 3 months. Available at: https://www.global-rates.com/en/interest-rates/euribor/2/euribor-interest-3-months/#:~:text=The 3-month Euribor interest rate is the interest rate,a maturity of 3 months. [date accessed 01st May 2024].

G.ogundur. (2020). Term deposit opening decision/classification problem. Available at: [date accessed 04th May 2024].

M,Hammadi. (2022). Identifying prospective clients for long term bank deposit. Available at: https://repository.rit.edu/cgi/viewcontent.cgi?article=12514&context=theses [date accessed 01st May 2024].